Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Weight

Meaning in my words How would you explain weight to a friend?	**How can you tell?** When a book rests on a table, which object supplies its weight?
Example or sketch Using 10 N/kg, what arrow represents the weight of a 3 kg book?	**Non-example and why** A book slides down a ramp. Should its whole weight arrow point along the ramp?

Normal force

Meaning in my words How would you explain normal force to a friend?	**How can you tell?** A crate rests on a ramp. Which way does the ramp push the crate?
Example or sketch How would you draw the normal force on a book on a level table?	**Non-example and why** Someone draws the normal force along the ramp. What would you rotate?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Tension

Meaning in my words How would you explain tension to a friend?	**How can you tell?** Which way does a slanted taut rope pull on a crate tied to its end?
Example or sketch Draw the force from a vertical cable on a hanging lamp.	**Non-example and why** Could a loose rope push a cart forward when you move its free end toward the cart?

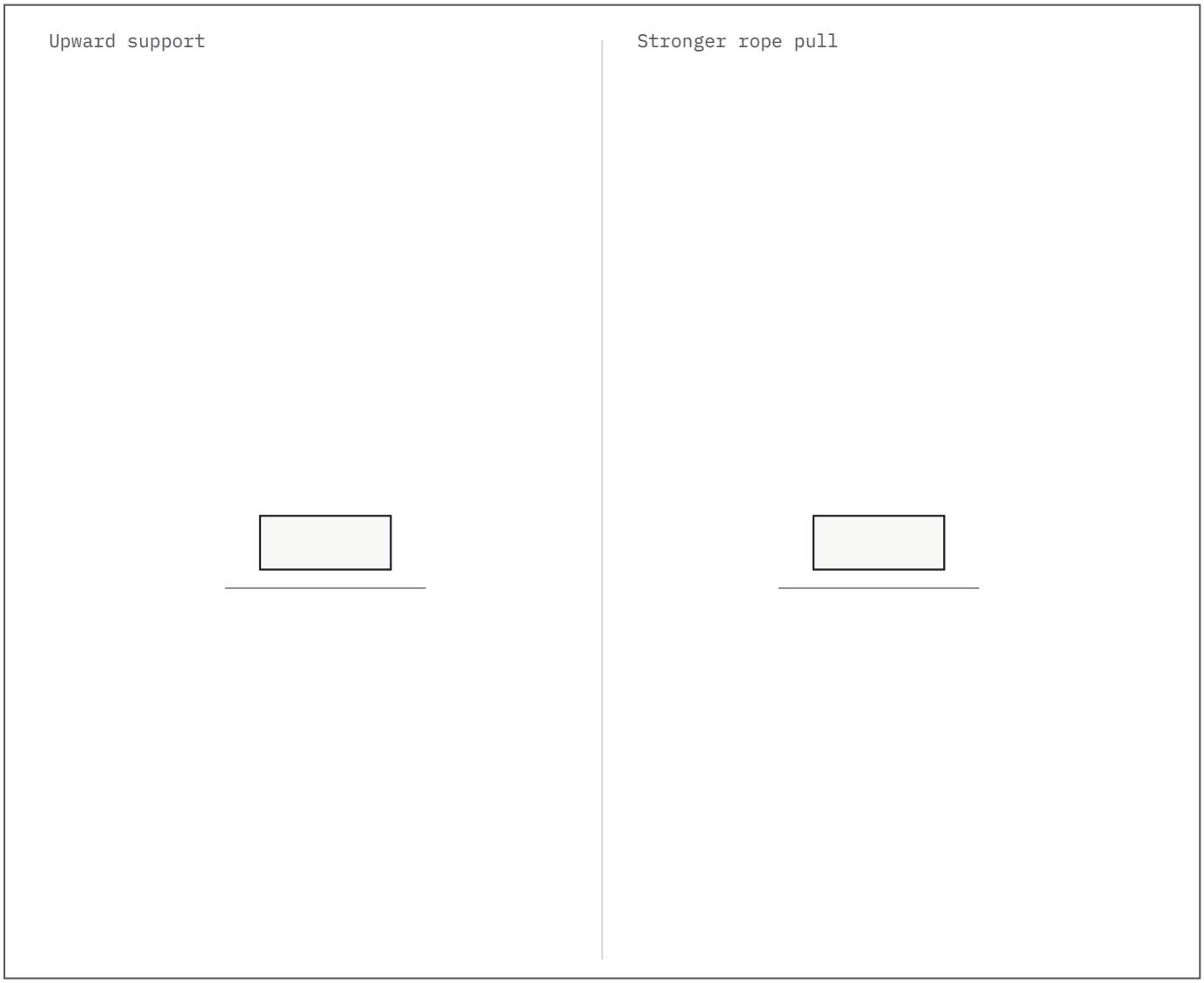
Receiver

Meaning in my words How would you explain receiver to a friend?	**How can you tell?** In “Earth pulls on the crate,” which object receives this force?
Example or sketch The rope pulls up on a lamp. On which object’s diagram do you draw that arrow?	**Non-example and why** “The cart moves right.” Can you name a force receiver from that statement alone?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a supported crate before and after an upward rope pull. Use the arrow lengths to show how the table's force changes.



My marks mean

On assignment: Receiver

SECTION E / BACK / FINAL MEETING ONLY Name _____

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Simulation engineer**

Prepare the force layer for XJ. Name a source and receiver for each arrow you keep. Put motion on a separate layer.

☐ **Science graphics editor**

Sketch the graphic viewers will see. Have your producer challenge the source of each arrow before you approve its label.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor’s check: Identify another real source that would require its own force arrow.